Experiment No: 05
Experiment Name:

1. Write an assembly language program to print letters A to Z using EMU8086.
2. Write an assembly language program to display a row of 80 stars using EMU8086.
3. Write an assembly language program to print a 5 × 5 matrix using EMU8086.

Theory: EMU8086 assembly handles repetitive output using loop instructions combined with DOS interrupts. To print letters A to Z, load 26 into CX as the counter, set AH to 2, place 'A' in DL, and call INT 21H inside a loop that increments DL each iteration. To display 80 stars, load 80 into CX, set AH to 2, place '*' in DL, and call INT 21H repeatedly until CX reaches zero. To print a 5×5 matrix, define a string with five stars and a newline terminator in the data segment, load 5 into CX, set AH to 9, point DX to the string, and call INT 21H inside a loop. The loop instruction automatically decrements CX and jumps back until the count finishes, making repetition efficient. Strings for function 09H must end with '$' to signal termination to the DOS interrupt handler. Finally, load 4CH into AH and call INT 21H to terminate the program and return control to DOS.

Code & Output:

i) Code to print letters A to Z:

```
.model small
.stack 100h
.code

main proc
    mov cx,26
    mov ah,2
    mov dl, 'A'

    loop1:
    int 21h
    inc dl
    loop loop1:

    exit:
    mov ah,4ch
    int 21h
    main endp
end main
```

Machine Code: B9 1A 00 B4 02 B2 41 CD 21 FE C2 E2 FA B4 4C CD 21

Output:



ii) Code to display a row of 80 stars:

```
.model small
.stack 100h
.code

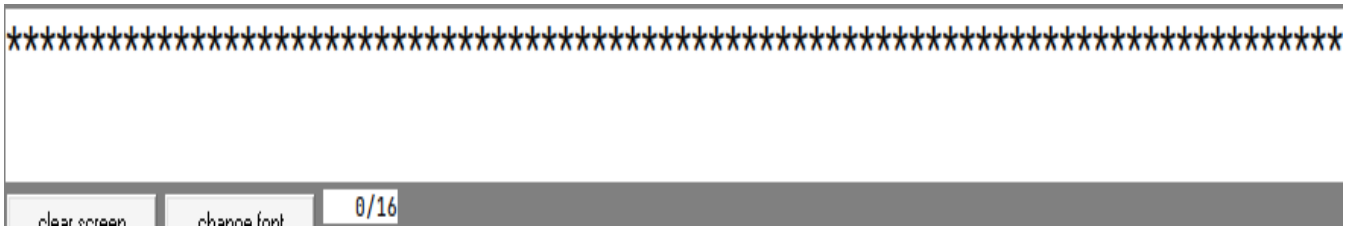
main proc
    mov cx,80
    mov ah,2
    mov dl, '*'

    loop2:
    int 21h
    loop loop2:

    exit:
    mov ah,4ch
    int 21h
    main endp
end main
```

Machine Code: B9 50 00 B4 02 B2 2A CD 21 E2 FC B4 4C CD 21

Output:

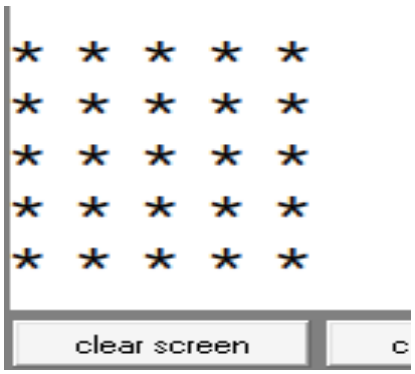


iii) Code to print a 5 × 5 matrix:

```
01 .model small
02 .stack 100h
03
04 .data
05 a db 10,13, '* * * * * $'
06
07 .code
08
09 main proc
10     mov ax,@data
11     mov ds,ax
12
13     mov cx,5
14
15     mov ah,9
16     lea dx,a
17
18     loop2:
19     int 21h
20     loop loop2:
21
22     exit:
23     mov ah,4ch
24     int 21h
25     main endp
26 end main
27
```

Machine Code: B8 20 07 8E D8 B9 05 00 B4 09 BA 00 00 CD 21 E2 FC B4 4C CD 21

Output:



Conclusion:

The program was executed successfully in the EMU8086 environment, and no errors were found. We successfully learned 3 assembly language program respectively print letters A to Z, display a row of 80 stars and print a 5 × 5 matrix.We also learned a new method of printing new line in assembly language program.